Economics S5040: Dynamic Stochastic Modeling
Lecture 11 — The Solow Growth Model

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1 Where We Are Going

For the last two weeks, our main object has been a value function. This week we temporarily step away from optimization and study one of the simplest dynamic models in macroeconomics:

$$\boxed{\text{How does capital accumulation determine long-run output?}}$$

Keep one assumption in mind throughout:

$$\boxed{\text{The saving rate is exogenous in the Solow model.}}$$

Households do not optimize over saving. We specify the fraction of output saved. In Week 7, saving will become a choice.

2 Production

Let

$$Y_t = F(K_t, L_t),$$

with constant returns to scale. Define

$$k_t = K_t/L_t, \qquad y_t = Y_t/L_t.$$

Then

$$y_t = f(k_t).$$

Our main example is

$$\boxed{y_t = Ak_t^{\alpha}}, \qquad A > 0, \quad 0 < \alpha < 1.$$

The marginal product of capital is positive,

$$f'(k) = \alpha A k^{\alpha-1} > 0,$$

but diminishing,

$$f''(k) = \alpha(\alpha-1)Ak^{\alpha-2} < 0.$$

3 Saving, Investment, and Depreciation

Suppose a fixed fraction s of output is saved:

$$S_t = sY_t.$$

In a closed economy, $I_t = S_t$, so investment per worker is $sf(k_t)$. If capital depreciates at rate δ and population is constant,

$$K_{t+1} = (1-\delta)K_t + I_t.$$

Hence

$$\boxed{k_{t+1} = sf(k_t) + (1-\delta)k_t.}$$

With Cobb–Douglas,

$$\boxed{k_{t+1} = sAk_t^{\alpha} + (1-\delta)k_t.}$$

Define

$$g(k) = sAk^{\alpha} + (1-\delta)k,$$

so $k_{t+1} = g(k_t)$. The state is k_t. Notice what is missing: there is no control chosen by a household.

4 The Steady State

A steady state reproduces itself:

$$k^* = g(k^*).$$

Therefore

$$\delta k^* = sA(k^*)^{\alpha},$$

or

$$\boxed{sf(k^*) = \delta k^*.}$$

Economically,

$$\boxed{\text{investment} = \text{replacement of depreciated capital.}}$$

For Cobb–Douglas,

$$\boxed{k^* = \left(\frac{sA}{\delta}\right)^{1/(1-\alpha)}}.$$

Steady-state output and consumption are

$$y^* = A(k^*)^{\alpha}, \qquad c^* = (1-s)y^*.$$

5 Why Does the Economy Converge?

Rewrite the law of motion:

$$k_{t+1} - k_t = sf(k_t) - \delta k_t.$$

Thus

$$sf(k_t) > \delta k_t \Rightarrow k_{t+1} > k_t,$$

and

$$sf(k_t) < \delta k_t \Rightarrow k_{t+1} < k_t.$$

Hence

$$k_t < k^* \Rightarrow k_{t+1} > k_t, \qquad k_t > k^* \Rightarrow k_{t+1} < k_t.$$

Diminishing marginal product is the key economic force.

6 The 45-Degree Diagram

Plot $k_{t+1} = g(k_t)$ against k_t and add the 45-degree line $k_{t+1} = k_t$. Their positive intersection is k^*.

- $g(k) > k$: capital rises.
- $g(k) < k$: capital falls.
- $g(k) = k$: capital is stationary.

The steady state is both a fixed point and the long-run destination.

7 Comparative Statics

From

$$k^* = \left(\frac{sA}{\delta}\right)^{1/(1-\alpha)},$$

we have

$$s \uparrow \Rightarrow k^* \uparrow, \qquad A \uparrow \Rightarrow k^* \uparrow, \qquad \delta \uparrow \Rightarrow k^* \downarrow .$$

Each result has a direct capital-accumulation interpretation.

8 Does More Saving Always Make Us Better Off?

A higher s raises k^* and y^*, but

$$c^* = (1 - s)f(k^*).$$

More saving raises future productive capacity while directly reducing the share of output consumed. Hence more saving need not raise steady-state consumption.

For Cobb–Douglas production, the Golden Rule saving rate is

$$\boxed{s_{\text{GR}} = \alpha.}$$

But Solow does not explain why households would choose it.

9 Competitive Factor Prices

With

$$Y = AK^{\alpha}L^{1-\alpha},$$

competitive factors are paid marginal products. The net return on capital is

$$\boxed{r = \alpha A k^{\alpha-1} - \delta},$$

and the wage is

$$\boxed{w = (1 - \alpha)Ak^{\alpha}}.$$

Thus

$$k \uparrow \Rightarrow w \uparrow, \qquad r \downarrow .$$

10 From a Model to Data

So far A, α, s, δ have just been parameters. For the United States we can ask:

- What is the capital share α?
- What fraction of output is invested?
- How rapidly does capital depreciate?
- What are K/Y and K/L?
- What value of A makes the production function fit U.S. output?

This turns Solow into a quantitative model.

11 Calibration

Calibration means

$$\boxed{\text{Choose model parameters to match selected features of the data.}}$$

Under competitive Cobb–Douglas production,

$$1 - \alpha = \text{labor share.}$$

Once Y, K, L, α are measured,

$$\boxed{A_t = \frac{Y_t}{K_t^{\alpha} L_t^{1-\alpha}}.}$$

This is the beginning of growth accounting.

Next lecture we will use the Penn World Table to construct these objects for the United States.

12 What to Remember

1. Output is produced from capital and labor.
2. A fixed fraction s of output is saved and invested.
3. Capital obeys $k_{t+1} = sf(k_t) + (1 - \delta)k_t$.
4. The positive steady state satisfies $sf(k^*) = \delta k^*$.
5. Diminishing returns generate convergence.
6. Higher saving raises steady-state capital but need not raise consumption.
7. Competitive factor prices are marginal products.
8. In Solow, saving is exogenous.
9. The model's parameters can be disciplined by actual data.